12

Radial Velocities

12.1 INTRODUCTION

The component of the relative velocity between the Earth and an astronomical object along the line of sight is called the radial velocity of that object. Since the wavelengths of lines are changed by such a velocity, as described by the Doppler shift, it is a relatively easy parameter to measure, and the earliest determinations date back to Huggins' visual spectroscopy (chapter 1). The Doppler formula gives the radial velocity from the change in wavelength (or frequency), with the convention that velocities away from the Earth are positive:

$$\frac{\Delta\lambda}{\lambda} = -\frac{\Delta\nu}{\nu} = \frac{v}{c} \tag{12.1}$$

where λ and ν are the rest wavelength and frequency of the line respectively, $\Delta\lambda$ is the wavelength change ($\Delta\lambda = \lambda_0 - \lambda$), $\Delta\nu$ is the frequency change ($\Delta\nu = \nu_0 - \nu$), and λ_0 and ν_0 are the observed wavelength and frequency of the line respectively, i.e.

$$v = \frac{\lambda_0 - \lambda}{\lambda}c = \frac{\nu - \nu_0}{\nu}c. \tag{12.2}$$

Equations (12.1) and (12.2) are applicable to velocities below about 10 000 km s^{-1}. As the velocities become an appreciable fraction of the speed of light, however, the relativistic version of the Doppler formula must be used:

$$\frac{\Delta\lambda}{\lambda} = -\frac{\Delta\nu}{\nu} = \frac{1 + (v\cos\theta/c)}{\sqrt{1 - (v^2/c^2)}} = z \tag{12.3}$$

where θ is the angle between the line of sight and the relative velocity vector, and z is the symbol conventionally used for redshift at high recessional velocities. We thus have for pure radial velocity ($\theta = 0°$):

$$v = \left(\frac{\lambda_0^2 - \lambda^2}{\lambda_0^2 + \lambda^2}\right)c = \left(\frac{(2\nu - \nu_0)^2 - \nu^2}{(2\nu - \nu_0)^2 + \nu^2}\right)c. \tag{12.4}$$

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Normally the Doppler effect only allows the radial velocity to be determined, but at relativistic velocities, the $\cos\theta$ term in equation (12.3) means that there is a wavelength change even for velocities directly across the line of sight. The transverse Doppler effect is obtained by setting $\theta = 90°$, and is given by

$$z_T = \frac{\lambda_0 - \lambda}{\lambda} = \frac{\nu - \nu_0}{\nu} = \frac{1}{\sqrt{1 - (v^2/c^2)}} - 1. \tag{12.5}$$

As can be seen from equation (12.5), the transverse shift is always to longer wavelengths. The transverse Doppler effect is rarely encountered; the peculiar object SS433 is the best known example, with its jets moving nearly across the line of sight at about a quarter of the velocity of light.

Heliocentric radial velocities are usually quoted. That is, the object's velocity with the line of sight component of the Earth's orbital motion removed. The heliocentric velocity (v_H) is related to the measured or geocentric velocity (v_G) by

$$v_H = v_G + v_E\,[\sin\delta_A \sin\delta_O + \cos\delta_A \cos\delta_O \cos(\alpha_O - \alpha_A)] \tag{12.6}$$

where v_E is the Earth's orbital velocity (29.78 km s^{-1} on average) α_O and δ_O are the right ascension and declination of the object, and α_A and δ_A are the right ascension and declination of the apex of the Earth's orbital motion at the time of the observation. They are given by

$$\delta_A = \sin^{-1}(\cot\delta_s \sin\alpha_s \sin\varepsilon) \tag{12.7}$$

$$\alpha_A = \alpha_s - \cos^{-1}(-\tan\delta_A \tan\delta_s) \tag{12.8}$$

where ε is the obliquity of the ecliptic (23° 27') and α_s and δ_s are the right ascension and declination of the Sun at the time of the observation.

12.2 TRADITIONAL APPROACH TO RADIAL VELOCITY DETERMINATION

The traditional approach to measuring astronomical radial velocities is from a photographic spectrogram which has a wavelength comparison spectrum (figure 9.6). As a rough rule of thumb, the radial velocity may be found to the same level of accuracy in km s^{-1} as the dispersion of the spectrum in Å mm^{-1} (i.e. $\pm$10 km s^{-1} from a spectrum with a dispersion of 10 Å mm^{-1}, or 1 nm mm^{-1}). The procedure is time-consuming (and very boring!). The steps in the process are as follows.

1. Identify the lines in the comparison spectrum and the object's spectrum (see below). Suitable tables of wavelengths are listed in appendix C. The comparison wavelengths, however, will normally be known from previous work.

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The more intense lines in the object's spectrum may usually be identified without difficulty unless the radial velocity is very large indeed (as for example with Quasars). The fainter lines may only be identifiable once an initial estimate of the radial velocity has been obtained and their corrected wavelengths found. Since the fainter lines will usually be narrower, and can be measured with a higher precision than the stronger lines, it is often necessary to approach the radial velocity by successive approximations.

2. Place the spectrum onto a measuring machine (figure 9.7). These machines take a variety of forms, but are all essentially high precision travelling microscopes which enable the positions of features in the spectrum to be measured to an accuracy of about one micron. Usually the microscope is fixed and the spectrum placed on a table which can be moved underneath it. The motion of the table is controlled by a precision screw thread from which its position may be determined by a Vernier indicator. The spectrum is aligned with the direction of motion of the table, and the measuring cross wires in the microscope eyepiece aligned parallel with the spectral lines (which are not always precisely perpendicular to the length of the spectrum).

3. Measure all the desired lines in both the comparison spectrum and the object's spectrum in a single pass down the spectrum, taking care to keep the screw moving in a single direction to avoid problems with backlash.

4. Turn the spectrum through 180°, re-align, and repeat the measurements for all the lines.

5. Return the spectrum to its original orientation and repeat steps 2, 3 and 4 as required.

6. Calling the measurements (or their averages) taken with the spectrum in its first orientation, x, and those with it reversed, y, a check on the accuracy of the results may be obtained by adding the x and y for each line together. The result of this addition should be a constant, representing the separation of the zero points for the measurements in the two orientations (figure 12.1). The scatter of the results of this addition enables an estimate of the accuracy of the measurements to be obtained. Any individual results which diverge significantly from the average are probably due to one or more mis-measurements of the line involved, and the data for that line have to be discarded.

7. The overall average of both the x and y measures may be obtained by subtracting the one from the other (there is no need to divide by the total number of measurements, since this just represents a scaling factor). The value of $x - y$ then represents the position of the line with respect to a point midway between the zeros for the x and y measurements (figure 12.1), which we may call m.

8. The first approximation to the relationship between position along the spectrum and wavelength may now be found. For a grating-based spectrum, the relation is approximately linear (equation (8.14)), and so we may use an equation of the form

$$\lambda = A + Bm \tag{12.9}$$

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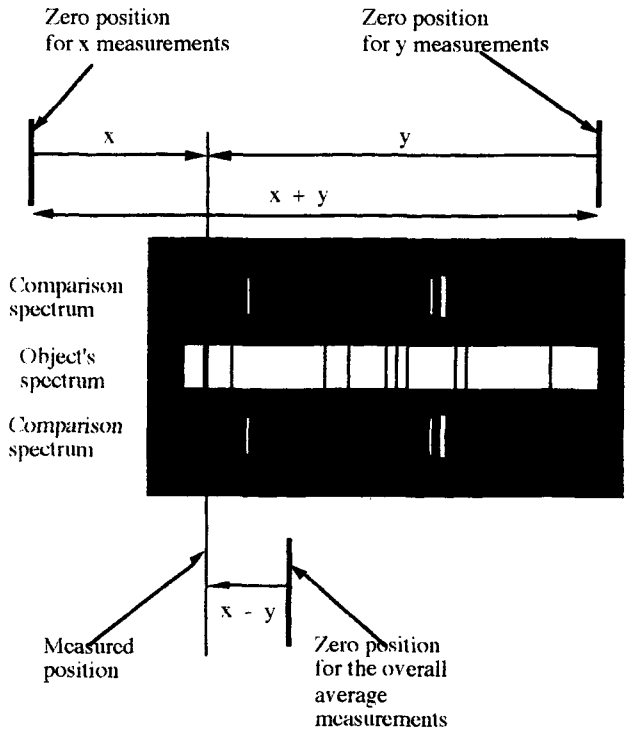

Figure 12.1 The x and y measurements of a spectrum for radial velocity.

where A and B are constants. For a prism-based spectrum, the relationship is nonlinear (equation (8.31)), and is more accurately represented by

$$\lambda = A + \frac{B}{m - C} \tag{12.10}$$

where A, B and C are constants. For a grating spectrum therefore, two comparison lines are selected about a third of the way in from each end of the spectrum. Using their measurements and known wavelengths, equation (12.9) is solved simultaneously to obtain values of A and B. For a prism-based spectrum three comparison lines are chosen, and equation (12.10) solved for A, B and C.

9. Equations (12.9) and (12.10) are only approximations. A more accurate relationship between measurement and wavelength has therefore to be found by constructing a correction curve. The correction curve is obtained by using the measurements for all the comparison lines, and determining measured wavelengths for them using either equation (12.9) or (12.10) as appropriate. This measured wavelength is then subtracted from the actual wavelength for each comparison line, and the difference (usually small) plotted against the actual wavelength to produce the correction curve.

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10. The observed wavelengths for the lines in the object's spectrum may now be determined. The initial approximation is obtained using the value for m for each line substituted into equation (12.9) or (12.10). These are then corrected for the approximate nature of the formulae using the correction curve.

11. The radial velocity of the object may now be found via the Doppler formula and the rest wavelengths for the lines in the object's spectrum.

12. If some lines in the object's spectrum could not previously be identified (but have been measured), their observed wavelengths may now be corrected for the radial velocity shift. The more precise values for their rest wavelengths may then allow identifications (see later) to be made. If need be, a second approximation to the radial velocity may then be found by returning to the previous stage.

13. The final value obtained for the radial velocity is then corrected for the Earth's motion (equation (12.6)) to give the heliocentric radial velocity.

12.2.1 Shortcuts

The traditional approach to determining radial velocities outlined above is clearly lengthy, especially if the measurement stages are repeated a number of times. Various approaches have therefore been tried to reduce the labour involved. However, for the highest precision work there is not much that can be done, though the strain on the operator can be reduced by using a TV camera on the microscope and working from the display on the monitor. A rather more advanced approach scans a short section of the spectrum and displays it both as a direct tracing and a reversed tracing. The measurement is made by superimposing the two tracings of a line. This approach also reduces the problem of measuring asymmetrical lines since it is easier to judge when two such profiles are aligned than to try to set a cross wire on the 'centre of gravity' of a single line. Electronic read-outs for the spectral line positions can be fed directly into a small computer at the press of a button by the operator, speeding up the whole process and eliminating mis-readings of the Vernier settings. Once the data are in the computer, then stages 6–13 can all be processed there as well.

If the radial velocity is required to less accuracy, perhaps to provide approximate rest wavelengths for unidentified lines, then several short cuts are possible. Firstly, simply fewer lines may be measured, with only one or two measurement sequences, for both the comparison spectrum and the object's spectrum. Secondly, if two comparison lines are close to a line from the object, then its wavelength may be found by straightforward interpolation. Thirdly, tracings of the spectra may be made and the wavelengths measured from them. For this to work, either the microdensitometer must be of a design which allows several tracings to be superimposed, without slippage, so that the comparison and object spectra may be traced separately, or the microdensitometer slit must be lengthened so that both the comparison lines and the object's lines are covered in a single tracing (figure 12.2). This last approach reduces the contrast

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considerably because of the greatly increased background coming through the long slit, and so fainter lines may not be distinguishable.

Some measuring machines scan the entire image and reduce it to a machine-readable form. In such cases, the data may be processed as discussed for directly produced machine-readable spectra below. Most such measuring machines (COSMOS, APM etc), however, are intended for scanning large plates from Schmidt cameras rather than the small plates commonly produced by photographic spectroscopes.

12.3 OBJECTIVE PRISM SPECTRA

The objective prism spectroscope (chapter 8) is highly efficient, both because of its high throughput, and because of the large number of objects simultaneously observed. Accurate radial velocities, however, cannot be obtained from objective prism spectra because they are usually of low dispersion, and it is not possible to provide a good comparison spectrum. One way around the problem, which will provide low accuracy data, is to take two exposures on the same plate, with the objective prism rotated through 180° between the two exposures (figure 8.24). If two or three or more stars with known radial velocities are on the plate, then the separation of a line in the two spectra may be calibrated in terms of velocity, and the velocities of the other stars determined.

Another approach is to use lines from terrestrial sources. The Earth's atmosphere produces absorption lines in the red and infrared parts of the spectrum; unfortunately this is the lowest dispersion part of an objective prism spectrum. A filter containing neodymium chloride has a single sharp absorption feature at 427 nm. Though the wavelength of the feature is temperature dependent, these filters have been used with some success.

For very low precision work, such as quasar identification, where velocities of 100 000 km s^{-1} or more need only be estimated to $\pm 10\%$ in order to provide candidates for more detailed study, then the cut-off in sensitivity of the photographic emulsion can be used. For some emulsions this can be quite sharp, giving a wavelength marker with an accuracy of 10 or 20 nm.

Treanor's method does not use an objective prism, but it is sufficiently closely related in principle to be dealt with here. The wavelength marker in the spectrum is a direct image of the object. Treanor's prism is a direct-vision prism placed in the collimated beam of light from a telescope. The prism is physically smaller than the beam of light, so that the spectrum produced by the prism has a direct image superimposed upon it from the light passing around the prism (figure 12.3). Like reversing an objective prism, this system also requires some stars with known radial velocities to be on the image in order to provide a calibration.

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12.4 MACHINE-READABLE SPECTRA

Many spectroscopes (chapter 10) now use detectors such as CCDs, IPCS, SEC Vidicon TVs etc, which have outputs fed directly into a computer. The principle of the determination of radial velocities from such spectra is the same as that for the traditional approach (above), but most of the labour is now undertaken by the computer. Line identification for the object's spectrum still usually has to be undertaken 'by hand' unless numerous very similar spectra are being processed. The comparison lines, however, will normally be the same from one spectrum to the next, and so their data only need to be provided once. The positions of lines are determined by fitting an appropriately shaped profile, or by the determination of the position of the 'centre of gravity'. Steps 8 and 9 of the traditional approach may simply be transferred to the computer, or more accurate formulae than equations (12.9) and (12.10) used from the start. Most modern spectroscopes will have appropriate software already written and readily available, so that the observer is hardly aware of the details of the process, but just obtains the radial velocities as his or her 'raw' data directly.

12.5 GRIFFIN'S METHOD

A ingenious direct reading radial velocity apparatus is due to Griffin. This employs a spectroscope of fairly conventional design, but places a previously obtained negative image of a spectrum at the focus. The transmitted light is then re-imaged onto a point source detector, and the total output measured. The negative spectrum must be from a similar type of object to that being observed (e.g. an A5 stellar spectrum could be used for stars say in the range A3 to A7). The negative image is moved along the real image, and when the two coincide there is a sharp drop in the output from the detector. This is because the brightest parts of the real spectrum will then coincide with the darkest parts of the negative, and the darkest parts of the real spectrum with the most transparent parts of the negative, whereas in other positions some of the bright parts of the real image will come through the transparent parts of the negative. The displacement of the negative can then be calibrated to read the object's velocity directly. The apparatus is particularly suited to the study of spectroscopic binary stars, where many radial velocity measurements of the same object are required.

12.6 LINE IDENTIFICATION

Most of the time, the identity of the main lines in a spectrum will be known from previous work by other astronomers (figure 11.2, for example). The fainter lines may also be known, but are more likely to be unidentified at the start of

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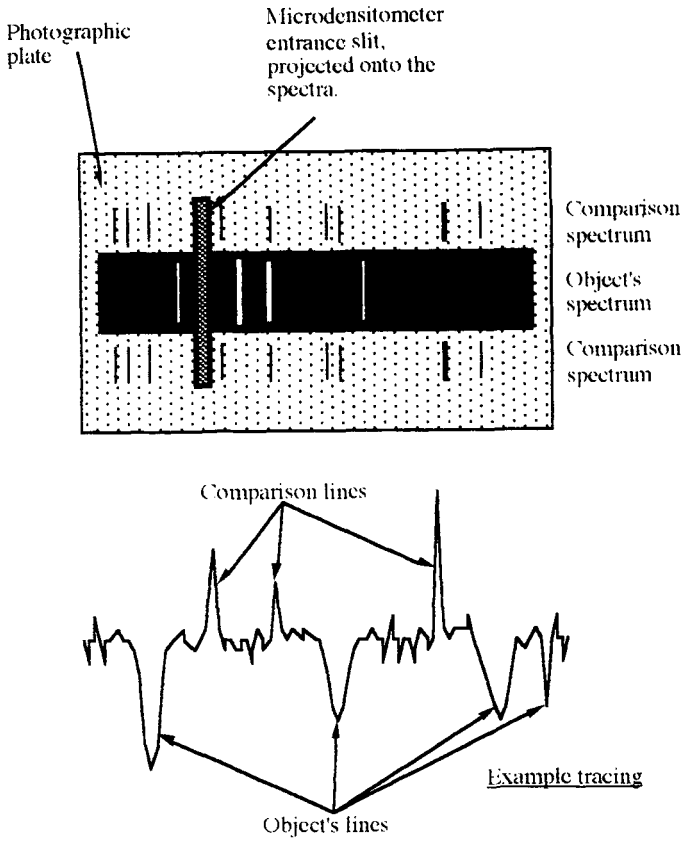

Figure 12.2 Using a long slit on a microdensitometer to obtain the comparison lines and the object's lines on a single tracing (spectrum shown as a photographic negative).

work on a spectrum. The observer is therefore faced with the problem of determining which elements, and in what levels of ionization and excitation, are producing these lines. Line identification can never be wholly certain, and there may remain unidentified lines, or lines with ambiguous identifications, no matter how much effort is put into the identification process. The following procedure, however, is likely to lead to reasonable success in identification.

1. Decide which features in the spectrum are lines and which noise. This is best done by comparing two or more spectra to find which features repeat (lines) and which do not (noise).
2. Measure the wavelengths of all features estimated to be lines (see earlier).
3. From the literature or any other sources, identify as many lines as possible.
4. Determine the radial velocity of the object, and correct all the lines' wavelengths to their rest values.

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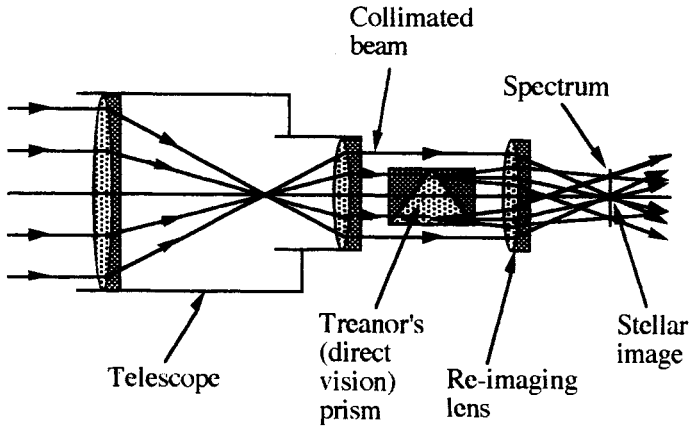

Figure 12.3 Treanor's approach to radial velocity determination.

5. From a set of tables of multiplets and their wavelengths (appendix C) find the atoms and ions from which lines have already been identified in the spectrum. List the other lines over the spectral region covered by the spectrum that come from those same atoms and ions. Check these lines against the wavelengths of the unidentified lines in the spectrum. If any of the values agree to within the uncertainty of the wavelength determinations, then there is a good chance that the unknown line has been correctly identified. Especial attention should be paid to resonance lines (lines originating from the ground state) and lines from low multiplets since these are likely to be relatively strong. Relative intensities within multiplets (chapter 4) can also be used as a cross-check. Thus if an identification has suggested one of the weaker lines in a multiplet, but the stronger lines from that multiplet are not found, then the original identification is probably wrong. On the other hand if several or all the lines of a multiplet are found and in about their correct relative intensities (remember the rules giving intensities within multiplets are very approximate), then this is as close to a certain identification of those lines as you are likely to get.

6. Repeat step 5 for the same atoms and ions but for one stage of ionization either side of those so far identified in the spectrum.

7. Return to step 4, if by now a significantly more precise value for the radial velocity may be found.

8. Check any remaining unidentified lines against a catalogue of line wavelengths (appendix C), and list all the lines to be found within the measurement uncertainty limits for the observed lines. This can sometimes be a large number of lines.

9. Eliminate improbable lines from the lists. This is a matter of judgement, and may well be wrong the first time round. It requires assessment of many items of information. Thus an estimate of the likely temperature of the object

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should be obtainable from those lines already identified. Molecular bands are then unlikely candidates for an object which contains lines from doubly or triply ionized atoms, or lines due to C IV in a spectrum known to contain lines of neutral metals etc. The normal abundances of elements can also be used. Thus, although a few stars do have strong lines of europium or praseodymium etc, the observer is unlikely to make many errors in rejecting them as initial candidates for identifications. Forbidden lines should not be expected unless dealing with interstellar nebulae. Lines from high multiplet numbers or intercombination lines are on the whole less likely than lines from lower multiplet numbers or from normal transitions, and so on.

10. Look for any repetitions amongst the remaining lines in the lists (i.e. two or more lines from the same element, same ion, same multiplet etc). With these potential identifications return to step 5. If there are no repetitions, or when they have been exhausted, return individually to step 5 for each potential identification for each unknown line. At this stage, or earlier, it may have become clear that some features initially judged as noise should now be taken as lines, or vice versa, and a return to step 1 is indicated.

11. If, after several iterations, some lines remain unidentified, then return to step 9 and consider progressively less probable identifications. If the object is an interstellar gas cloud, or could include rarefied gases in an outer shell, then forbidden lines from metastable levels will also need to be considered.

12. Finally, consider that some lines may be shifted by a different radial velocity from that of the object; for example, lines from the interstellar medium or from a companion in a binary system. In such cases the lines are likely to be representative of quite different levels of excitation and ionization from the lines from the object itself. There are also symbiotic stars wherein lines from apparently different conditions do coexist.

13. Unfortunately when all the above has failed, you can no longer invent a new element! Though this was an option open to Norman Lockyer who discovered helium in this manner in 1868. Any remaining unidentified lines nowadays must remain so.

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